Homework 4

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Problem 1. After the lecture on Monday, I was asked if there are languages A and B which are both non-regular, but AUB is regular. The answer is that there are such languages. Find and explain an example of such A and B. Hint: Try AUB = Σ *.

Proof. To prove that:

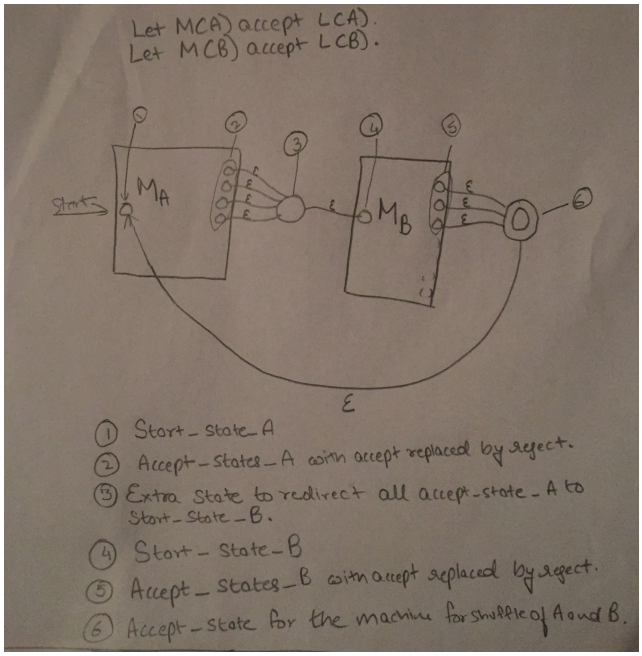
There is a non-regular language A and a non-regular language B such that AUB is a regular language.
Proof:

$\Sigma^* = \{0,1,\epsilon\}$
Let A = {w| w∈ (0^i 1^j) s.t. i >=j and i>=0 and j >=0 }
We proved this to be non-regular in class.
Let B = {w| w∈ (0^i 1^j) s.t. j >=i and i >= 0 and j>=0 }
This is non-regular with the same logic as proved in class.
AUB = {w| w∈ (0^i 1^j) s.t. j >=i and i >=j and i and j are greater than equal to 0 }
Therefore, we can see that the union is just {w| w ∈ $0^m 1^n$ where m and n can be anything}
This evaluates to Σ^*.
A DFA for this should be fairly easy to make.

□

Problem 2. For languages A and B,the shuffle of A and B is the language:SH=w:w=a1b1a2b2...akbk, where a1...akA and b1...bkB.That is, string w in SH can be decomposed into strings (which might be longer than just a single character) that alternate between being in A and being in B. Show that the class of regular languages is closed under shuffle.That is, if A and B are regular languages, then so is the shuffle of A and B.Hint: Non-determinism helps here.

Proof. **1 NFA**



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Since, we can make a NFA for it, and we know that the language accepted by a NFA is closed. Thus, we know that the language of shuffle of two regular languages is closed. □

Problem 3. In class, we proved that the language E= $\{0^i 1^j$:i> j$\}$is not regular. We used the string $0^{p+1}1^p$. Would the proof work if we had chosen S = $0^{p+2}1^p$ Explain. Would it work if we had chosen S = $0^{p+2}1^{p+1}$. Explain.

Proof. **For string $\mathbf{0^{p+2}1^p}$**
This will not work, since we can have a loop of one zero, which would be in the language, by pumping down, the initial condition will hold true; hence, it will hold our assumption.

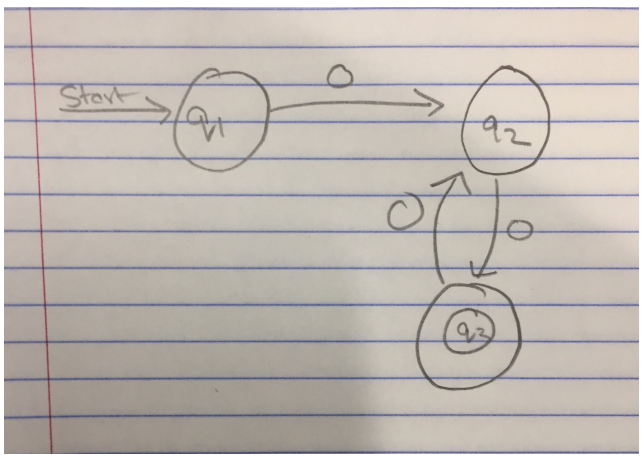
For string $\mathbf{0^{p+2}1^{p+1}}$
This will work, since by pumping down worst cast i.e. single 0, it will work.

□

Problem 4. For a string w, define w^r to be the string w in reverse order. For example, if w= abc, then w^r=cba. Define the language P(Σ) as {ww^r w$\in\Sigma$}.Each string $ww^r$is called a palindrome

Problem 4a. Show that if Σ ={0}, then P(Σ) is regular.

Proof. **2 DFA**



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We can make a DFA for this, since this will be accepted by strings of even lengths only as the character is always going to be 0. Therefore,as we can make a DFA for this language, it is a regular language. □

Problem 4b. Then show that if is any alphabet with more than one character,then P (Σ) is not regular. To show this, write out a proof using the template proof approach. That is, dont just refer to the Pumping lemma.

Proof. **To prove that**

If Σ is any alphabet,more than one letter, P(Σ) will not be regular.

Proof:

Let P(Σ) be regular.

Let m be the # of states in the machine M that accepts P(Σ).

Lets take a string w, $0^{m+1}1^m$ s.t. ww^r $\in$ P(Σ).

ww^r = $0^{m+1}1^m1^m0^{m+1}$

Therefore, we are guaranteed a loop in 0s and not in 1 for w.

let a_k be the first state that is visited twice in w.

Now, for string S, ww^r first part of w is visited before a_k, second part of w between the first and second visited, and the remaining part is visited after the second visit. Lets call the first part X, second part Y, and the third part be Z.

X = 000...0000

Y = 0000....0000

Z = 000111..111w^r

Lets consider the string S'= XYYZ. Since, Y is processed during a traversal of string back to a_k. XYYZ would be the same as XYZ and the machine M will accept it.

Since it has more characters in w than in w^r the midpoint will shift to the left. However, at this point the number of 1s on the right will be more than the number of 1s on the left of the midpoint. This cannot happen since in a palindrome, the left and right side should have equal number of 1s. Here, we run into a contradiction since, w' is accepted by the machine, but S' is not in P(Σ). So M does not exist and P(Σ) is not regular.

□

Problem 4c. Try to make the template proof approach work to prove that $P(\Sigma)$ is not regular when $\Sigma = \{0\}$. Of course, since we know that this $P(\Sigma)$ is regular, the proof should fail. What specifically goes wrong in that proof attempt?

Proof. This proof will follow the same template as the previous problem; however, the conclusion will not hold true, since as we have only one character, even after the midpoint shifts, the number of a particular character isn't guaranteed to be different as we only have one character $\{0\}$. $\square$